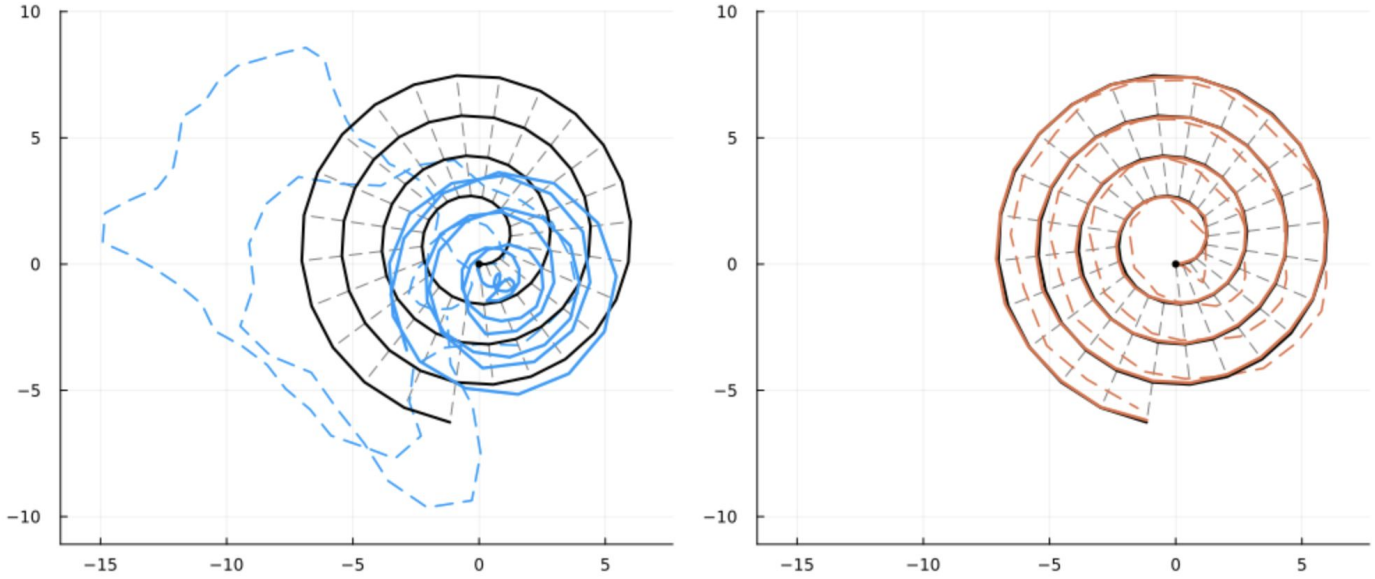
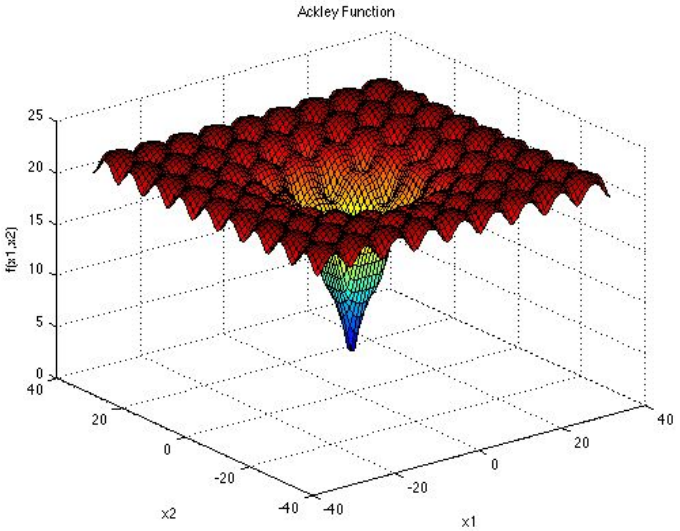
Spectral Initialization for 2D Pose Graph SLAM

MATH7207 Project 1 | Clayton Schmidter



What is (Spectral) Initialization?

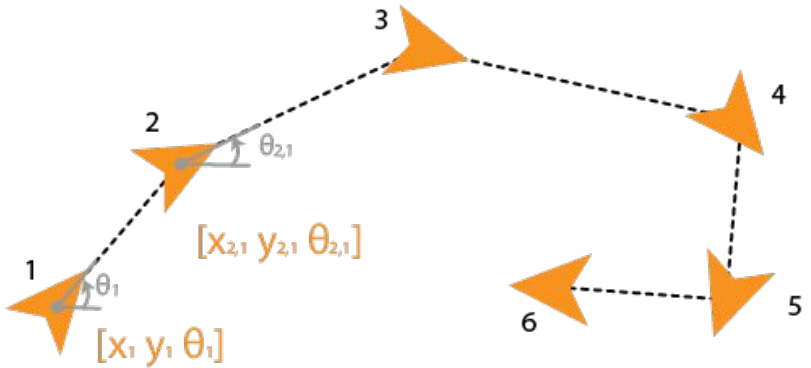
- The process of finding a good initial guess
- Crucial for minimizing a nonconvex function with local solvers (LM, GN, etc.)
- A local solver usually only finds the bottom of whatever basin it starts in
- Bad start -> local minimum; good start -> (hopefully) global minimum
- “Spectral” = using eigenvectors



2D Pose Graph SLAM

- Each pose is a position and a heading $X_i = (x_i,\, y_i,\, \theta_i) \in SE(2)$
- Each edge (i, j) measures where pose j is as seen from pose i
  - Odometry edges: consecutive poses (i -> i+1)
  - Loop closures: revisiting a place seen earlier
- Independent Gaussian error on each edge:

$$Z_{ij} = X_i^{-1} X_j \,\mathrm{Exp}(\varepsilon_{ij}), \qquad \varepsilon_{ij} \sim \mathcal{N}(0, \Sigma_{ij})$$



- Gaussian MLE = weighted least squares:

$$\hat{X} = \underset{X_2,\ldots,X_n}{\arg\min} \sum_{(i,j)} r_{ij}^{\top} \Omega_{ij}\, r_{ij}, \qquad r_{ij} = \mathrm{Log}\left(Z_{ij}^{-1} X_i^{-1} X_j\right)$$

Solved with LM, but where do we start?

Baseline: Dead Reckoning

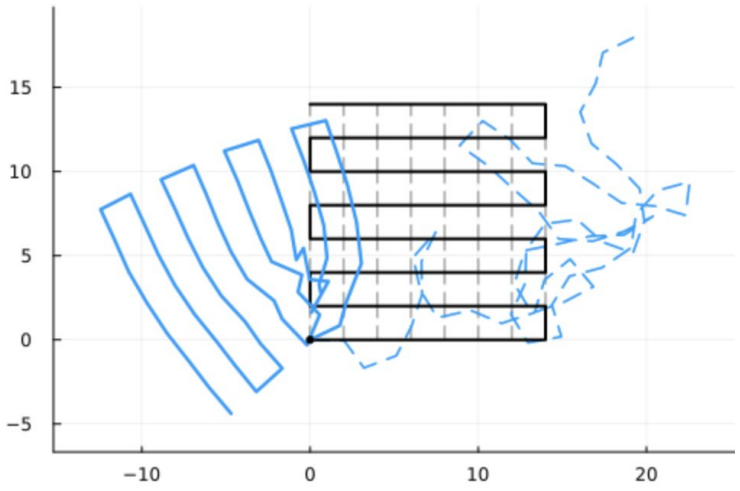
Chain the odometry from pose 1 and ignore loop closures

$$X_1 = I, \qquad X_{i+1} = X_i \, Z_{i,i+1}$$

Heading error accumulates:

$$\hat{\theta}_k = \theta_k + \sum_{m=1}^{k-1} \varepsilon_m^{\theta}$$

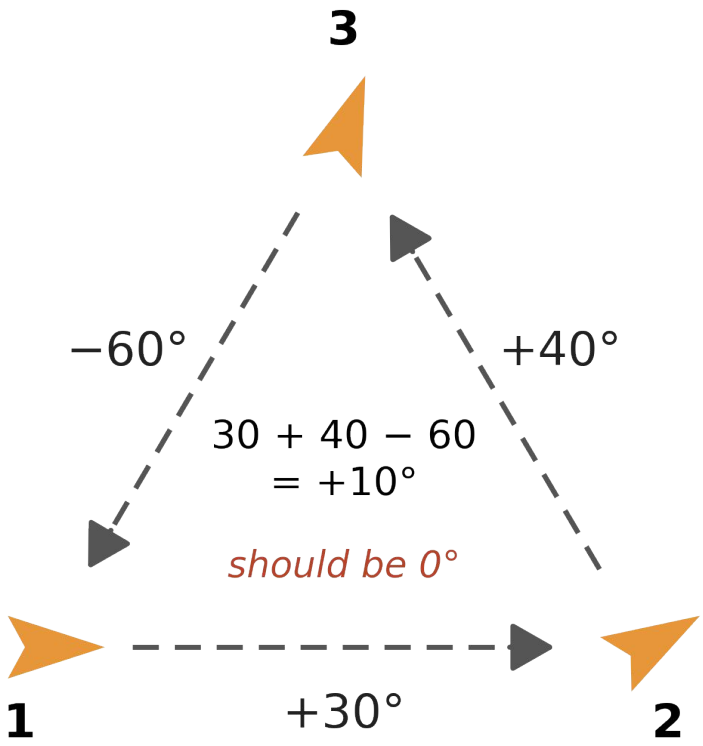
One bad heading early on bends the rest of the trajectory



Dashed = dead reckoning start
Solid = after LM (stuck).

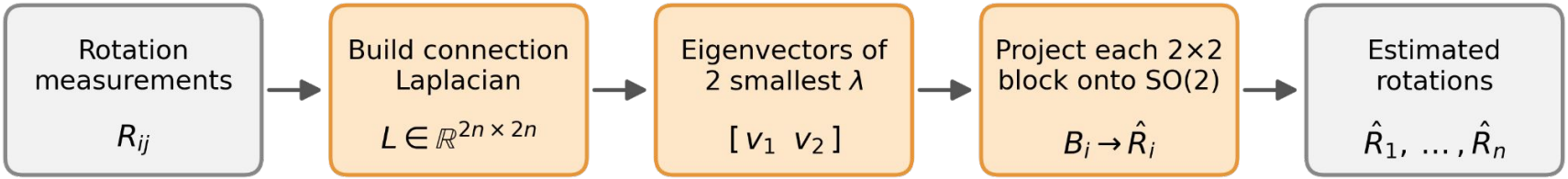
Spectral Synchronization

- Nonconvexity is in the rotations. Translations are linear once rotations are known
- Each edge measures a heading difference, and these should sum to 0 around a loop
- Dead reckoning puts all the error on one edge, and the best answer spreads it out
- Headings as points on a circle -> quadratic -> eigenvector problem



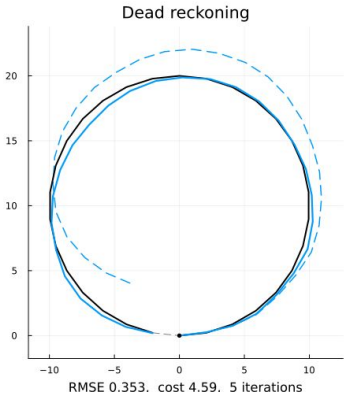
The Spectral Step

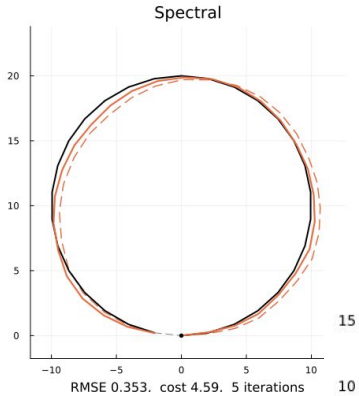
- L = connection Laplacian, built from the rotation measurements. It measures how inconsistent the rotations are (zero = perfectly consistent).
- No noise -> the true rotations are L's null space
- Then: translations by linear least squares



Spectral Initialization in Practice

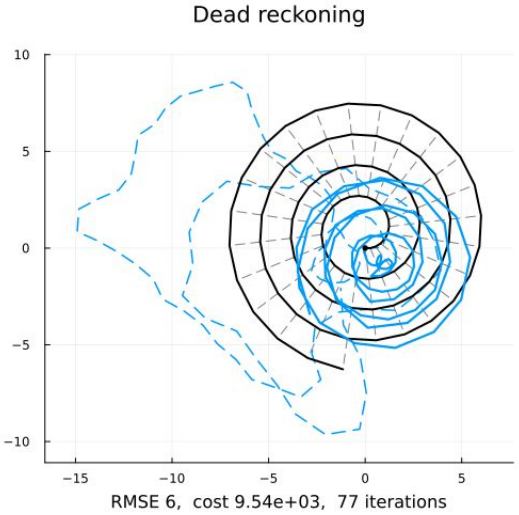
Moderate noise

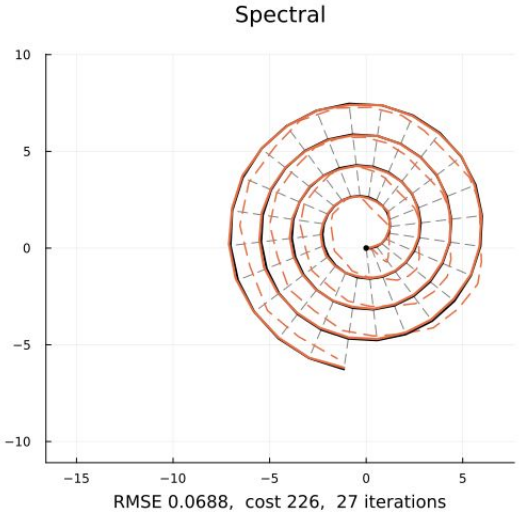


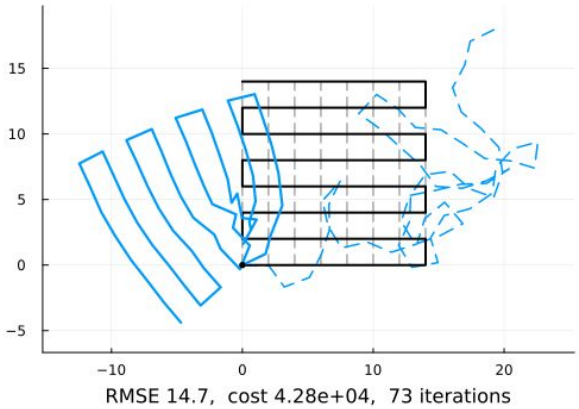


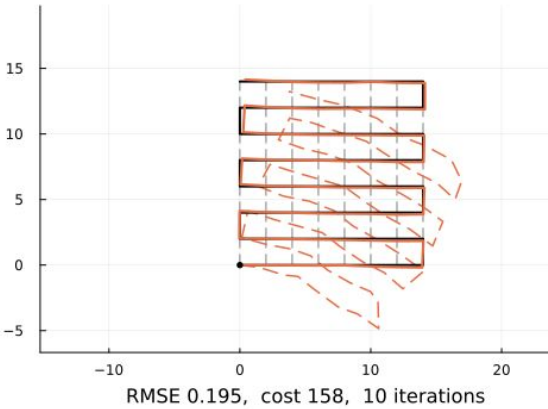
Dashed = start · Solid = after LM ·
Black = truth · Gray = loop closures

High heading noise









Takeaways and What's Next

- **A good initial guess decides whether the solver finds the global minimum or gets stuck in a local one**
- This was an "easy" case: a compact group (SO(2)) and Gaussian noise
  - Noncompact groups (SE(d)) -> nonsymmetric matrices
  - Non-Gaussian noise or outliers -> least squares is no longer the right objective
- "Looks good" isn't proof: LM finds a minimum, not necessarily the global minimum
  - Certifiable methods can prove global optimality

Singer (2011) · Arrigoni et al. (2016) · Rosen et al. (2019)